



SYMBIOSIS
CENTRE FOR INFORMATION TECHNOLOGY

T01 – Sales Performance Dashboard App

MBA – Data Sciences and Data Analytics

(MBA-DSDA Batch 2026–2028)

SALES PERFORMANCE DASHBOARD APPLICATION

T01 Assignment Report

Python and Streamlit based interactive dashboard for sales performance analysis

Student Name: Samyak Acharya

PRN: 26030242002

COURSE: AI Powered Developer Platform & Tools

1. Introduction

For this assignment, I developed a comprehensive Sales Performance Dashboard using the T01 dataset. The main objective was to create an interactive, executive-level dashboard that helps Regional Sales Heads understand sales performance across multiple dimensions including time periods, regions, channels, categories, and products.

I built a professional Streamlit application where users can apply dynamic filters and view key performance indicators (KPIs) such as total revenue, units sold, average order value, and gross margin percentage. The dashboard also includes advanced features like month-over-month growth analysis, cross-filtering capabilities, and automatically generated business insights.

The project was developed in Python using VS Code, with Streamlit for the web interface, Pandas for data manipulation, OpenPyXL for Excel file handling, and Plotly for interactive visualizations. The application includes both light and dark theme support for enhanced user experience.

2. Objectives

- Build a working dashboard for analyzing sales performance from the T01 dataset
- Calculate key business metrics including revenue, units, AOV, and gross margin percentage
- Implement interactive filtering across months, regions, channels, and categories
- Create dynamic visualizations that respond to filter selections
- Generate month-over-month growth analysis with proper handling of sequential data
- Develop business insights that are data-driven and contextually relevant
- Implement professional UI/UX with theme toggle and responsive design
- Ensure data validation and robust error handling throughout the application
- Document the AI-assisted development process and lessons learned

3. Dataset Used

I used the Sales sheet from the provided T01_Sales_performance_dashboard_app.xlsx file. The dataset contains approximately 2,000 transaction records with comprehensive sales information including order details, geographic data, channel information, product categories, pricing, and profitability metrics.

Key columns analyzed:

- order_id - Unique transaction identifier
- order_date - Transaction date (with timezone handling)
- region, city, state, city_tier - Geographic segmentation
- channel - Sales channel (Online Store, Retail, etc.)
- category, product - Product classification and identification
- units, unit_price_inr - Quantity and pricing information
- discount_pct - Applied discounts
- revenue_inr, cost_inr, gross_margin_inr - Financial metrics

Before performing calculations, the application validates data integrity, handles date parsing (including timezone strings), and ensures numeric data types are properly formatted.

4. Tools and Technologies Used

Tool	Purpose	How I Used It
Python 3+	Core programming language	Main development environment for all logic
Streamlit	Web application framework	Interactive dashboard interface and widgets
Pandas	Data manipulation	Excel file reading, data filtering, and aggregations
OpenPyXL	Excel file handling	Reading .xlsx files with multiple sheets
Plotly	Interactive visualizations	Charts, graphs, and interactive elements
VS Code	Development environment	Code writing, debugging, and project management

5. Key Performance Indicators (KPIs)

5.1 KPI Definitions and Formulas

KPI	Formula	Business Purpose
Total Revenue	SUM(revenue_inr)	Overall sales performance
Units Sold	SUM(units)	Volume measurement
Average Order Value	Total Revenue / Unique Orders	Customer spend analysis
Gross Margin %	SUM(gross_margin_inr) / SUM(revenue_inr) × 100	Profitability analysis

5.2 Advanced Calculations

Month-over-Month Growth:
MoM Growth % = ((Current Month Revenue - Previous Month Revenue) / Previous Month Revenue) × 100

- Top Product Analysis: Revenue-based ranking with dynamic filtering support
- Regional Performance: Cross-dimensional analysis with proper aggregation

6. Core Features Implemented

6.1 Interactive Filtering System

The dashboard implements a comprehensive filtering system with Apply/Reset button functionality. Users can filter data across four dimensions: Month, Region, Channel, and Category.

The system uses Streamlit session state to manage temporary filter selections and applied filters separately, preventing performance issues from real-time filtering on large datasets.

- Multi-select dropdowns for each filter dimension
- Apply button mechanism to batch filter changes
- Reset functionality to clear all filters
- Visual feedback showing active vs pending filters

6.2 Executive Dashboard Layout

Professional executive-level dashboard layout with KPI cards, interactive visualizations, and business insights. The design follows dashboard best practices with clear information hierarchy and responsive design.

- KPI card layout with formatted currency and numbers
- Monthly revenue trend with month-over-month growth calculations
- Multi-dimensional analysis views (Region, Channel, Category, Products)
- Automated business insights generation
- Professional styling with proper spacing and typography

7. Technical Implementation Challenges

7.1 Data Processing and Validation

The T01_Sales_performance_dashboard_app.xlsx dataset required careful handling of mixed data types, date parsing with timezone information, and currency formatting. I implemented robust data validation to ensure calculations are accurate and handle edge cases like empty filters or missing data.

- Implemented pandas data type conversion with error handling
- Created date parsing logic to handle timezone strings in order_date column
- Added data validation functions to verify KPI calculation accuracy
- Built error handling for file loading and sheet access
- Ensured proper handling of null values and data integrity

7.2 Streamlit State Management

Managing filter states and theme preferences required careful implementation of Streamlit session state. The challenge was preventing unwanted re-runs while maintaining responsive user interaction.

- Implemented dual-state system (temp_filters vs applied_filters)
- Added proper state initialization with default values
- Managed component keys to prevent state conflicts
- Optimized re-run triggers for better performance

8. Validation and Quality Assurance

8.1 KPI Validation Benchmarks

All KPI calculations were validated against known benchmarks to ensure accuracy. I established validation targets and implemented automated testing to verify calculation integrity.

KPI	Expected Value	Validation Status
Total Revenue	₹24,092,037.95	✓ Validated
Units Sold	12,001 units	✓ Validated
Average Order Value	₹12,046.02	✓ Validated
Gross Margin %	33.39%	✓ Validated

8.2 Functional Testing

Comprehensive testing was performed across all filter combinations, theme modes, and edge cases to ensure robust application behavior.

- Filter functionality: All combinations of month, region, channel, and category filters
- Theme switching: Complete UI consistency in both light and dark modes
- Data integrity: Verification of calculations across different filter states
- Error handling: Graceful handling of missing files and invalid data
- Performance testing: Response times for large filter operations
- Cross-browser compatibility: Testing across different browser environments

8.3 Code Quality Standards

The codebase follows Python best practices with proper documentation, error handling, and modular architecture suitable for enterprise deployment.

- PEP 8 compliance for Python code formatting
- Comprehensive docstrings for all functions
- Type hints where applicable for better code maintenance
- Error handling with informative user messages
- Separation of concerns with modular utils package
- Performance optimization with Streamlit caching decorators

9. Business Insights Generated

9.1 Automated Insight Engine

The dashboard includes an AI-powered insight generation system that analyzes filtered data and provides contextual business observations. These insights adapt dynamically based on user selections.

9.2 Key Business Findings

Based on the T01 dataset analysis, several key business patterns emerged:

- Revenue Performance: Total revenue of ₹24+ million across all channels and regions
- Order Volume: 12,001 units sold with healthy average order value of ₹12,046
- Profitability: Strong gross margin of 33.39% indicating healthy business model
- Growth Trends: Month-over-month analysis reveals seasonal patterns and growth opportunities
- Regional Performance: Geographic analysis shows performance variations across different markets
- Channel Effectiveness: Multi-channel analysis reveals optimal sales channel performance

9.3 Strategic Recommendations

- Focus on high-performing regions and channels for expansion opportunities
- Investigate seasonal trends for better inventory and campaign planning
- Optimize product mix based on category performance analysis
- Leverage month-over-month growth data for forecasting and budgeting

10. AI-Assisted Development Process

10.1 Development Methodology

This project utilized AI-powered development tools within the Kiro IDE, demonstrating modern human-AI collaborative software development. The process combined AI assistance with human oversight for optimal results.

10.2 AI Contributions

- Code Architecture: AI suggested modular structure with utils package organization
- Feature Implementation: Assisted with complex Streamlit state management and CSS styling
- Problem Solving: Provided solutions for theme integration and filter optimization
- Documentation: Helped generate comprehensive technical documentation
- Testing Guidance: Suggested validation approaches and test scenarios
- Best Practices: Recommended industry standards for dashboard development

11. Responsible AI Practices

11.1 Ethical Development Approach

The project followed responsible AI development practices, ensuring transparency, accountability, and ethical use of AI assistance throughout the development lifecycle.

11.2 Transparency and Documentation

- Clear documentation of AI contributions vs human contributions
- Transparent reporting of development methodology and tool usage
- Comprehensive code comments explaining logic and decision rationale
- Open acknowledgment of AI assistance in project documentation

- Detailed testing and validation records for accountability

11.3 Quality Assurance and Bias Prevention

- Human validation of all business logic and calculations
- Cross-verification of data processing and aggregation methods
- Testing across diverse data scenarios to prevent algorithmic bias
- Regular review of AI suggestions for appropriateness and accuracy
- Implementation of robust error handling and data validation

12. Project Limitations

12.1 Current Scope Limitations

While the dashboard successfully meets the assignment requirements, several areas could be enhanced in future iterations:

- Data Source: Single Excel file dependency; production systems would require database integration
- Real-time Updates: Static data analysis; live business systems need real-time data feeds
- Advanced Analytics: Basic statistical analysis; could benefit from predictive modeling
- User Management: Single-user application; enterprise deployment needs authentication
- Scalability: Designed for dataset size in scope; larger datasets may require optimization

12.2 Technical Constraints

- Browser Dependency: Requires modern web browser with JavaScript enabled
- Python Environment: Needs specific Python version and package dependencies
- Memory Usage: Large datasets may require additional memory optimization
- Network Requirements: Web-based interface requires stable internet connection for deployment

13. Future Enhancement Opportunities

13.1 Advanced Analytics Features

- Predictive Analytics: Forecasting models for revenue and sales trends
- Machine Learning: Customer segmentation and behavior analysis
- Statistical Analysis: Correlation analysis and significance testing
- Advanced Visualizations: Geographic maps, network graphs, and advanced chart types

13.2 Enterprise Integration

- Database Integration: Connect to SQL databases and data warehouses
- API Development: REST APIs for programmatic access to dashboard functionality
- User Authentication: Role-based access control and user management

- Automated Reporting: Scheduled report generation and email distribution
- Mobile Optimization: Responsive design for tablet and mobile devices

14. Personal Learning Reflection

14.1 Technical Skills Development

- Streamlit Framework: Mastery of interactive web application development
- Data Visualization: Advanced Plotly charting and interactive visualization techniques
- Python Development: Enhanced skills in pandas, data manipulation, and modular programming
- UI/UX Design: Understanding of dashboard design principles and user experience
- State Management: Complex application state handling and performance optimization

14.2 Business Analysis Insights

The project deepened my understanding of business intelligence requirements and the importance of translating raw data into actionable insights for executive decision-making.

- KPI Selection: Understanding which metrics matter most for sales performance analysis
- Executive Reporting: Creating clear, concise visualizations for senior stakeholders
- Data Storytelling: Presenting data in ways that drive business understanding
- Filtering Strategy: Designing intuitive interfaces for complex data exploration

14.3 AI-Assisted Development Learning

Working with AI development tools provided valuable insights into modern software development practices and the future of human-AI collaboration in programming.

- Prompt Engineering: Effective communication with AI development assistants
- Code Review Skills: Critical evaluation of AI-generated code and suggestions
- Collaborative Development: Balancing AI assistance with human expertise and judgment
- Quality Assurance: Maintaining high standards while leveraging AI productivity gains

15. Conclusion

This Sales Performance Dashboard project successfully demonstrates the integration of technical skills, business acumen, and AI-assisted development practices. The application provides a comprehensive, interactive platform for sales analysis that meets executive-level reporting requirements while maintaining technical excellence.